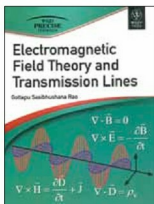


Electromagnetic Field Theory and Transmission Lines

This publication, which covers almost all the topics of electromagnetic field theory with relevant mathematical treatment, is a comprehensive text book for UG and PG students, as well as for research scholars.

Though electromagnetics involves mathematical analysis for better comprehension, the readers need not necessarily require a mathematical background as the text is easy-to-read, logical, and has ample number of illustrations, diagrams, appropriate worked examples, and relevant details wherever it is required. It helps the readers to get a good grasp of EMFTL and helps them realise the utility and indispensability of knowledge in this subject.

The book follows a step-by-step approach and all of its ten chapters begin with analysing theoretical basics and physical interpretations, and then progresses to an advanced level. It covers topics, like vector algebra, the fundamental laws (Coulomb's law and Gauss' law, laws of Biot-Savart and Ampere), Maxwell's equations, EM wave propagation in transmission lines, wave guides and free space.



By: G. Sathishkumar Rao
Published by: Wiley India Pvt Ltd,
4435-36/7, Ansari Road, Daryaganj,
New Delhi 110002
Pages: 410
Price: Not mentioned

Digital and Analog Communication Systems

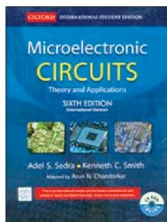
This text on communication theory has been written for undergraduates and first-year graduate students. It presents an introductory level treatment of digital and analogue communication systems with an emphasis on digital communication systems.

The book covers diverse aspects of communication systems from the mathematical and statistical abstraction of information theory, modulation theory, and coding theory to the electronic considerations associated with the building of functional blocks to perform various signal processing tasks. It also has a review section, which briefly discusses signal models, system analysis, random variables and random processes.

However, it is not suitable for those who have no background of the subject as, while writing the text, the author has assumed that the prospective readers already have some knowledge of circuit analysis and system analysis, and they also have some prior exposure to Fourier Transforms, random variables and random processes.



By: K. Sam Shanmugam
Published by: Wiley India Pvt Ltd,
4435-36/7, Ansari Road, Daryaganj,
New Delhi 110002
Pages: 600
Price: Not mentioned



By: Adel S. Sedra and Kenneth C. Smith
Published by: Oxford University Press,
VMCA Library Building, 1 Jai Singh Road,
New Delhi 110001
Pages: 1384; Price: ₹ 725

Microelectronic Circuits: Theory and Applications

The latest sixth international version of the book is meant for the students of electrical and computer engineering. Though it is a text for the core courses in electronic circuits, it is also useful for engineers and other professionals who want to update their knowledge of the subject.

This book is aimed at making the reader an expert in analysing and designing electronic circuits, both analogue and digital, discrete and integrated. While the application of integrated circuits is covered in the text, the main emphasis is on transistor circuit design. The prerequisite for studying the material in this book is a first course in circuit analysis. For review, some linear circuits material is included in the appendices—specifically, two-part network parameters in Appendix B; some useful network theorems in Appendix C; single-time-constant circuits in Appendix D; and s-domain analysis in Appendix E. The authors have assumed that the reader has no prior knowledge of physical electronics, thus all required semiconductor device physics is included and Appendix A provides a brief description of IC fabrication. All these appendices can also be found in the CD that accompanies this book.

The emphasis on design has been increased in this edition by including more design examples, simulation examples, exercise problems, and end-of-chapter problems. Also considerable material is provided on the most valuable design aid, SPICE.

In the new edition several changes have been made to both organisation and coverage of content. Some of the most important topics covered in this book are: MOSFETs and BJTs, semiconductors, op-amps, cascode configuration, SPICE, CMRR and CMOS.